

Distinguishing Drone Types Based on Acoustic Wave by IoT Device

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Abstract—A drone detection system is essential to protect people from misuses, infringement on privacy, and other harms by drones. In this paper, we investigate to distinguish drone types based on acoustic wave. Our proposed drone detection technique employs Dejavu that is an acoustic fingerprint identification tool and random forest to classify types of drones. In the study, we first collected a dataset for three types of drones: DJI Spark, Quadcopter AR Drone 2.0, and Parrot Mambo drone. Then, we compared the detection performance based on the dataset. The experimental results showed that each drone sound can be classified by random forest.

Keywords—Drone, UAV, IoT, Sound, Acoustic fingerprint, Random forest, Acoustic wave

I. INTRODUCTION

To safely fly an unmanned aerial vehicle also known as a drone, we need to understand the aviation law and ordinance for each country and pay attention to them [1]. At present, many types of drones are sold, and they are expected to be useful in various fields. On the other hand, drones may be used for wrongdoing. For instance, in 2015, police arrested a man who flew a drone carrying radioactive sand on to the top of the Japanese prime minister's office [2]. The incident showed that it could be easy to attack by using drones, so that an amendment to the aviation act came into effect on the 10th of December 2015 in Japan. After that, flying drones over residential areas without permission and during night time are prohibited. Also, drones in open airspace must stay below 150 meters and keep at a distance of at least 30 meters from people, properties on the ground, and vehicles [3].

From considering such incidents, drone detection is necessary to protect civilians from an abused drone. Recently researchers have been studying to detect drones, for instance, we have detection methods based on an acoustic signal, Wi-Fi statistical fingerprint analysis [4], and image processing. However, as the commercial products are expensive, it is impossible for an individual to install them.

In this paper, we propose a detection and distinguishing method using a machine learning algorithm (i.e., the random forest) to classify drone types based on a fingerprint identification concept by an acoustic wave. In our approach, we first collect drone sounds as a dataset and minimize the noises within it. We employ an IoT device as the detection system. Besides, we focus on detection and distinguishing hobby drones, DJI Spark, Quadcopter AR Drone 2.0, and Parrot Mambo drone with FPV camera, that may infringe privacy. As this paper tries to distinguish the type of each drone by using an acoustic wave, we provide a new concept to focus both on drone detection and distinguish the type of drone based on an acoustic wave in an uncontrolled

environment and we used a machine learning algorithm to classify the drone sound.

This paper is organized as follows: Section 2 surveys existing work for drone detection based on acoustic wave. Section 3 proposes our detection and distinguishing method, and Section 4 then shows experimental results. Finally, Section 5 provides concluding remarks.

II. RELATED WORK

This section surveys drone detection studies using acoustic wave. Reference [5] experimented with the drone detection based on the acoustic signature using a support vector machine (SVM) to recognize the drone audio fingerprint. Also, reference [8] reviewed techniques that rely on ambient radio frequency signals, radars, acoustic, and computer vision techniques for detection.

From the above existing studies, we avoid the bias as the same method in reference [5] that when they performed the experiment, the train classes included 868 segments as drone frames and 868 segments as crowded frames, so the sample of drone sounds and the sample of environment sounds were equally separated. Besides, we remove the background noise of all audio files before used in the experiment from the result of reference [7] that stated the unstructured noise performed inferior results.

Acoustic fingerprint has been used as a source localization in many pieces of the existing work. However, there are some experiments that their results are not as the expected because of the quality of the chosen microphone. For example, reference [6] summarized the quality of the microphone and the sampling frame size are also factoring that impacts the result. For a reason, we carefully chose the microphone to record the sound.

As described above, the concept of acoustic fingerprint identification has been used in some research for detecting drones. In this paper, we then focus on distinguishing the types of drones in particular.

III. PROPOSED METHOD

A. Microphone selection

There are three microphones contains a list of candidates: ConferenceCam CC3000e, Rerii-SF960B, and Dell 7559's microphone. We formulate the strategy to select the microphone by using Parrot Mambo drone as a test device, Audacity as audio recording software, and recorded in the controlled environment. It was decided that the best of these three microphones for this investigation is Rerii-SF960B. It can record the sound with the distance more than

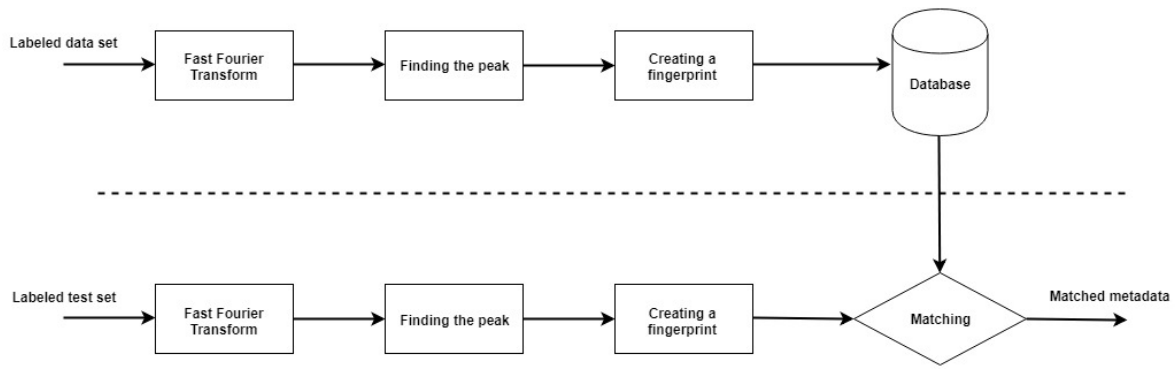


Fig. 1. An overview of the Dejavu process to identify the sound

30 meters. However, the strength of volume depends on the distance between the drone and the microphone which is lower when the distance is greater.

B. Data preprocessing

Audacity was chosen as an audio recording software for recording drone sounds. The recorded environment needs to be silenced, and minimized noises as much as possible. For this paper, drones sound was recorded in the hall room with a distance of 1, 3, and 5 meters between the microphone and each drone. Besides, the problem for recording drone sounds is each drone has an approximate battery life 5 – 10 minutes; thereby, it is necessary to change a battery every time that takes more time to finish the recording. In addition, it is essential to collect drone sounds for testing. The design of the strategy is to record in the uncontrolled environment which has many noises; for instance, birds chirping and human voices. The dataset contained three different drones: DJI Spark, Quadcopter AR Drone 2.0, and Parrot Mambo drone with FPV camera and each drone included four different distance which are 1, 3, 5, and 10 meters between each drone and the microphone.

Audacity provides Spectrogram view which uses the Fast Fourier Transform to represent the frequency of information versus time [9]. Besides, the purpose of converting is conveniently separated foreground sounds from background noises because of in the form of a sound wave, it is difficult to differentiate noises from the audio track, so in the form of a spectrogram, we can know which part of acoustic wave contained more or less energy.

Data cleansing is the method of correcting corrupt from a recorded set to identifying incomplete or incorrect parts of the data then replacing, modifying, or erasing the coarse data [9]. We thus decided to reduce the noise of all audio files to make the quality of sound purer as more as possible, so it was complete by Audacity. We assume that noises are corrupt and inaccurate which may make the detection result erroneously. To emphasize, the noise reduction algorithm uses Fourier analysis [10] in which the frequency spectrum of the sound found in the spectrum of pure tones that compound the background noise in the quiet sound segment.

Making a dataset is necessary to have the same pattern and format so that there are some libraries in Python that works involve with Audio such as Pydub. Each audio divided into the small part, the duration of each sample is approximately 4 to 5 seconds, and the lower than above was removed.

C. System design

Since the world becomes interconnected, IoT devices are increasingly extensive in a variety of industries. The concept of drone detection by using IoT devices is to be portable and can be remotely monitored and controlled. We used the Raspberry Pi 3 model B to connect with a microphone and a speaker. As a reason for selecting Raspberry Pi as a portable computer, the Raspberry Pi is low-priced and has many necessary features such as built-in Wi-Fi to send the data through the internet. Consequently, the recorded sound is sent from Raspberry Pi to the cloud server for computing the similarity. If the recorded sound is identified as drone sounds, the speaker makes an alarm to alert that there is a drone in the area.

D. Acoustic Fingerprint Identification

Dejavu is the recognition tool implemented in Python that contained many necessary algorithms for identifying the audio. The concept of Dejavu is to create the audio fingerprint of each audio which memorizes audio by listening to it once and fingerprinting it. Then by playing a song and recording microphone input, Dejavu attempts to match the audio against the fingerprints held in the database and returning the song being played [11]. Fig 1. described the process of Dejavu, starting with listening to an audio track, performing Fast Fourier Transform in overlapping windows over the length of the song, extracted peaks, and formed fingerprints. The purpose of using this tool is to observe that the acoustic fingerprint process can distinguish the type of drones by using an acoustic wave.

The Fast Fourier Transform (FFT) is an algorithm for calculating the Discrete Fourier Transform (DFT) of a sequence or its inverse which is more quickly and get the same result as DFT. Fast Fourier Transform can reduce the number of computations and capacity of storage, and convenient application in an instability test [12]. The FFT was repeatedly used to plot a spectrogram that takes the audio in blocks of size samples.

Finding peaks in each sample in a spectrogram regarding amplitude defines a peak as a pair by time and frequency corresponding to an amplitude value, so it is the greatest in a local neighborhood around it. Other pairs around it are lower in amplitude, and therefore less likely to exist noise.

The hash function procures an integer input and returns another integer as an output. The created hash represents the unique fingerprint of audio. Each fingerprint hash value contains frequencies of peaks, the time difference between

peaks. Alternatively, more peaks in a fingerprint mean a rarer fingerprint that more strongly would identify the sound, but more peaks additionally mean less robust against noises.

E. Random forest

Random forest is a kind of machine learning algorithm for both classification and regression [13]. Random forest is an ensemble learning that performs by assembling a mass of decision trees at training time and outputting the class which is the mode of the classes or mean prediction of the individual trees [14]. In the field of acoustic, random forest is used to detect, classify, prediction, and others; for instance, using the random forest to classify based acoustic event detection [15].

IV. EXPERIMENT

The purpose of the experiment is to observe the performance of acoustic identification method to detect and distinguish the type of drone based on acoustic wave in an uncontrolled environment and combines with image detection, radiofrequency, temperature, and others to increase the accuracy of detection which helps to identify the objective and the ability in order to assess the harm that will be occurred from drones.

The experiment was conducted in two segments: before and after adding more dataset. For each segment of the experiment, it was tested about 15 times for each segment in different parameters because we need to find the suitable of the parameter to make the proper result. In details, there are three parameters that overlap ratio is the ratio by which each sequential window overlaps the last and the next window, fan value is a degree to which a fingerprint can be paired with its neighbors, and peak neighbor size is the number of cells around an amplitude peak in the spectrogram in order for Dejavu to consider it a spectral peak. In the first segment, the test set was contained three types of drone, four tracks for each type. The dataset was included drone sounds 630 samples, and environment sounds were selected only 630 samples to avoid bias as a result. The second segment was included the same test set but adding more dataset inside a database which contained drone sounds 1,940 samples and environment sound 1,940 samples. We performed an acoustic fingerprint identification in Python 2.7 with MySQLdb and Dejavu, and the Random forest in Python 2.7 with Scikit-learn, TensorFlow, and Librosa on the following system: Intel(R) Core (TM) i7-6700HQ CPU 2.60GHz CPU and NVIDIA GeForce GTX 960M.

A. Data set

The number of the audio files that were collected is total 64 files, 52 for the database, and 12 for the testing set. The total length for the dataset that kept in the database is 2 hours 40 minutes and 48 seconds, and 14 minutes and 1 second for the testing dataset. The dataset of database size is 1.62 Gigabytes, and 141.6 Megabytes for testing set.

B. Results

- Random forest

The result from the random forest represented in the confusion matrix. It indicated each drone sound was classified correctly as shown in Fig. 2.

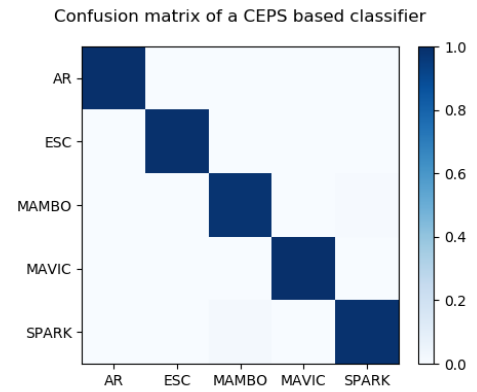


Fig. 2. The result of random forest

- Acoustic Fingerprint Identification

The dataset and test set are not the same track, so the detect rate depended on the result from the output of the first segment of the experiment. The data was collected from 240 records. The dataset contained 1,260 samples, and the test set contained 12 audio files.

TABLE I. THE RESULT OF THE FIRST SEGMENT

Drone type	Drone detected successfully rate	Distinguish drone type successfully rate
DJI Spark	96.67	90
AR Drone 2.0	58.33	25
Parrot Mambo	81.67	31.67

TABLE II. THE RESULT OF THE SECOND SEGMENT

Drone type	Drone detected successfully rate	Distinguish drone type successfully rate
DJI Spark	96.67	83.33
AR Drone 2.0	55	30
Parrot Mambo	60	53.33

The result from the output of the second segment of the experiment. The data was collected from 240 records. The dataset contained 3,880 samples, and the test set contained 12 audio files. From both results, DJI Spark has the highest rate in drone detection and distinguishing drone type.

V. CONCLUSION

In this paper, we investigated the performance of using acoustic fingerprint identification to detect and distinguish drone sounds, and the efficacy of random forest to classify the data. Each drone sound can be classified correctly using the random forest. However, the performance to identify the sound which produced by drones is different from songs in term of frequency was good in DJI Spark, but weak in Quadcopter AR Drone 2.0, and Parrot Mambo drone. Eventually, a high amount of data in the database also increases the distinguish rate that helps to identify the sound from each drone.

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